Lilly Sweet

(913) 335-2048 • lillysweet@berkeley.edu • LinkedIn • Portfolio

EDUCATION

University of California, Berkeley MS/PhD in Mechanical Engineering — Aug 2025 – Dec 2026

Graduate Coursework: Continuum Mechanics, Machine Learning, Rheology, Adv. CFD & Heat Transfer (I & II), Adv. Thermodynamics, Adv. Fluid Mechanics

Research: NSF Fellow | Formulated and solved ill-posed inverse problems in limited-angle X-ray PIV and energy resolved spectroscopy to reconstruct particle trajectories and metal alloy mass fractions.

University of Kansas BS in Mechanical Engineering — Jan 2021 – May 2025

GPA: 3.98 *(with distinction)* **Involvement:** Aero Co-Team Lead: Formula SAE

WORK EXPERIENCE

SpaceX Raptor Graduate Engineer Hawthorne, CA — May 2025 – Aug 2025

- Integrated an **AI camera system** to automate machining process inspections, eliminating human-eye subjectivity and achieving **5–20× faster** throughput with improved repeatability and reliability. Utilized RobotStudio and an **IRB 1300**.
- Implemented **automated data capture** and reporting for equipment process and part flow, eliminating manual tracking and enabling more accurate throughput analysis for the foundry.
- **Automated** ingestion and organization of **nozzle extension test data** into an MQTT-backed database, enabling engineers to rapidly query results and validate propulsion hardware quality.

(Incoming) Aero/Thermal Falcon Graduate Engineer Hawthorne, CA — May 2026 – Aug 2026

Tesla Controls Engineering Intern Fremont, CA — May 2024 – Aug 2024

- Integrated PLCs with Ignition-based SCADA/HMI systems, enabling plant-wide data visibility and centralized monitoring of critical manufacturing operations.
- Led a **PLC migration** effort covering >$200K in hardware and labor, minimizing downtime while modernizing controls infrastructure for long-term reliability.
- Participated in hands-on courses for KUKA and FANUC robotics systems as well as sought out opportunities to implement and apply those skills on the manufacturing line. Assisted teams with end-effector design.

Burns & McDonnell Mechanical Engineering Intern Kansas City, MO — May 2023 – Aug 2023

- Developed **automated engineering workflow tools** by combining Python and PowerShell scripting, enabling unattended execution of complex analysis routines previously performed manually or in Excel.
- Collaborated with the CFD team to integrate aerodynamic analysis into ductwork design for datacenters.

US Army Combat Medic Sergeant (M-Day) — Jan 2020 – Jan 2026

- **Security Clearance:** Secret **Rank:** Sergeant (E-5) (Non-Commissioned Officer) **Honorable Discharge**
- Served as co-NCOIC for joint Armenian and U.S. rotary wing MEDEVAC training; led mountain ruck-based mass casualty exercises and helicopter evacuation operations.

ENGINEERING PROJECTS

- **n^{th} Order FVM Euler Solver:** Used spectral, FR/CPR, and DG methods to achieve n^{th} order accuracy on 1D Euler flow through a converging/diverging nozzle.
- **2D Viscous with Unstructured Meshing FV Solver:** A custom solver that simulates supersonic flow over a NACA0012. Roe's flux, Superbee & MC limiters (2^{nd} order w/ LS) were able to successfully resolve shocks.
- **FSAE Aerodynamics:** Ran iterative STAR-CCM+ simulations. Improved the team's process for higher fidelity simulations. Mechanically designed and manufactured mounting and wings

SKILLS

- **Analysis:** STAR-CCM+, Patran/Nastran, Ansys (Mechanical & Fluent), Thermal Desktop
- **Coding:** Python, MATLAB, C++/Arduino, Machine Learning (TensorFlow, Keras, pandas)
- **CAD/General:** SolidWorks, NX, ABB Robot Studio, PLC's (ABB, Siemens), Ignition